

AI System Engineering Ethics

Case Study: The UK A-Level Algorithm (2020)

Ethical Failures in Algorithmic Decision-Making in Public Education

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Abstract:

The British government along with its educational regulatory body called Ofqual adopted an automatic grading algorithm for examinations in place of school tests due to the coronavirus outbreak. The system was made to stop "grade inflation" by comparing teacher-assessed grades to how well the school had done in the past. The model uses historical data from schools instead of individual students, which leads to systemic bias that unfairly lowers the grades of students from state schools and disadvantaged backgrounds while favouring smaller private schools. When the results came out, about 39.1% of the grades that teachers had predicted were lower, which caused a national outcry and many protests. This case study examines the ethical shortcomings of the Ofqual algorithm, emphasizing the consequences of insufficient transparency and the prioritization of statistical distribution over individual equity, which led to a complete reversal by the government. The algorithm was dropped in favor of original teacher assessments just a few days after the crisis.

1. Introduction

The A-Level (Advanced Level) tests are the most important high-stakes tests in the UK for getting into college and getting a good job in the future. COVID-19, which was unprecedented, affected school attendance throughout the country, causing the cancellation of the May/June exams. At the same time, the government requested from the regulator Ofqual to develop another method for awarding grades for all students so they could enter higher education. The new method no longer depended on the performance of each individual student but relied on an automated algorithm that "moderated" grades. This case study aims to analyze the motivations behind the decision, the way the algorithm worked, and the societal backlash towards its implementation.

2. Case Study Description

2.1 Impact of COVID-19 and Exam Cancellation:

Due to the requirements of social distancing, the outbreak of COVID-19 resulted in the impossibility of holding traditional examinations. On March 18, 2020, the UK government made it official that all GCSE and A-Level tests would not take place. It was asked that the teachers submit Centre Assessment Grades (CAGs), which were their estimates of how well their pupils would do, as well as a ranking list of their pupils according to their grades in the subjects they were studying.

2.2 The Justification Behind the Implementation of the Algorithm:

The Department for Education was concerned that using only CAGs would lead to grade inflation due to the fact that teachers tend to overestimate the marks their students could achieve.

As a result, Ofqual created a "standardisation algorithm" to make sure that the national grade distribution stayed the same as it had in the past. The algorithm primarily took into account:

- Historical Evidence: The success of the school in each subject area in the previous three years (2017-2019).
- Past Achievement: Achievement of the students in past national examinations (GCSEs).
- Class Sizes: The formula did not penalize groups containing fewer than 5-15 students, thus providing an advantage to private schools with smaller class sizes.

2.3 Sequence of Events (2020):

- March 20: Closure of schools; cancellation of the exams shortly thereafter.
- May 29: Deadline for teachers to submit CAGs and ranking of students.
- August 13: A-Level Results Day. Massive downgrading is revealed; 36% to 40% of grades are lower than teacher predictions.
- August 14–16: Widespread protests; legal challenges launched against Ofqual.
- August 17: The Government performs a "U-turn," announcing that students can keep their original CAGs if they were higher than the algorithm's grade.

2.4 Results and Public Backlash:

In addition, the publication of results exposed the existence of a “zip code lottery,” where high-performing students who had attended state schools which traditionally performed poorly found themselves with lower grades than deserved based on the distribution of results to create a grading curve. Poorer students experienced grade reductions of 10.4 percentage points, whereas wealthier students only received reductions of 8.3 points. The system was branded as “classist” and “biased,” eventually leading the government to discard the algorithmic approach.

3. Description of the AI System

3.1 Purpose and Design Rationale:

The main goal of the algorithm was to make sure A-Level grades in 2020 were similar to those in previous years. *Ofqual* worried that if teachers' grades were used without changes, then it would lead to higher grades than usual because teachers often give higher predictions. So, the algorithm was meant to fix this issue. However, the way it was set up caused unfair problems.

Note: Ofqual - Office of Qualifications and Examinations Regulation, England

3.2 System Inputs:

The algorithm used three main types of input data: (1) teacher predicted grades from the CAG process, (2) student rankings in their class, and (3) each school's grade history from the last three

years, broken down by subject. National data for each subject was also included to help compare schools.

Table 1

Input–Process–Output Structure of the Ofqual Grading Algorithm (2020)

INPUT	PROCESS	OUTPUT
Teacher Centre Assessed Grade (CAG)	Statistical standardisation weighted by school history	Final A-Level grade (A*–E or U)
Student rank within class cohort	Applied per subject, per school	University admissions decision
School 3-year historical grade distributions	Compared to national subject distributions	40% of grades lower than teacher prediction

Note: Compiled from Ofqual (2020) and Burgess and Sievertsen (2020).

3.3 Standardisation Process and Output:

The standardization process changed grades based on past school performance. Instead of just trusting teacher predictions, the model looked at how likely it was for a student to get the predicted grade based on the school's history. If a school usually had low high grades, the model would lower the predicted grades, even for the top students. The final grade for each student was sent to universities for admissions, making it a very important and permanent decision.

The results were clear. At *private schools*, where students usually did better, teacher predictions were more likely to be *accepted*. At *state schools* in poorer areas, students were more likely to get *lower grades*. This was a result of how the model was set up, not a mistake.

4. Where Problems Began: The System Lifecycle

4.1 Data Selection

The main design choice was using past school performance as a key factor. This data was not fair. It showed years of inequality in the UK education system. Schools in poor areas often got lower grades because they had fewer resources and faced social challenges, not because students were less capable.

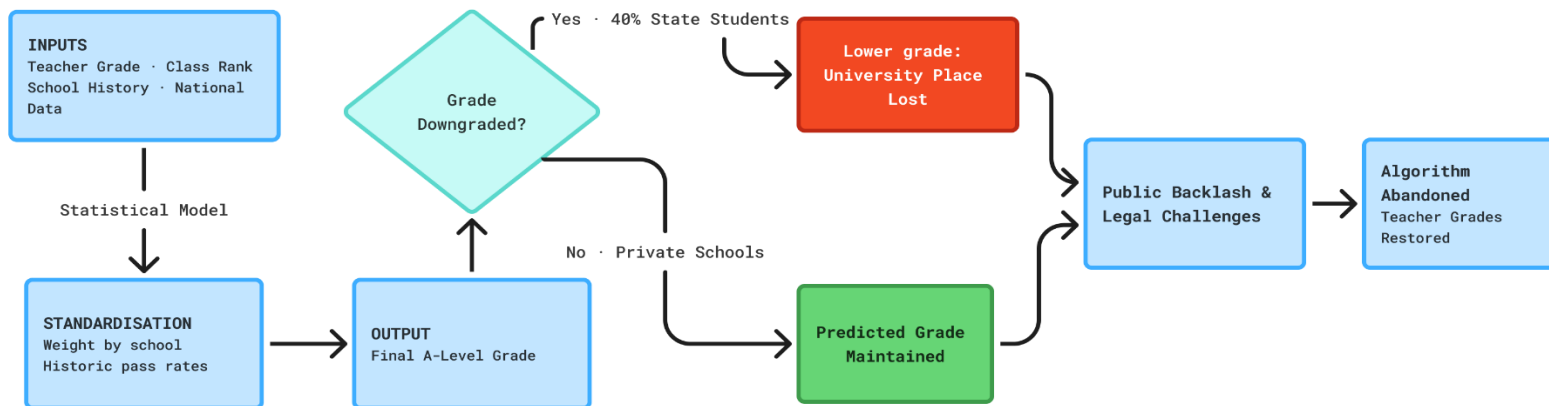
By using this past data to judge individual students, the system continued existing unfairness and bias. Also, looking at just three years of data was not enough for small groups of students (sometimes only 5 or 6). This led to unreliable grade results for the schools that needed help the most.

4.2 Modelling Approach

A second big problem came from using a group model for individual students. The system was made to show grade trends for groups, not to predict how well single students would do. A student who had worked hard and improved a lot, or who had done better than others despite a tough background, was judged not on their own progress but on the overall results of all students from their school in the last three years. This is a basic mistake in how the model was used.

4.3 Transparency and Accountability

The algorithm process could not be known before the outcome was revealed. Neither the students nor the educators knew how the historical information was applied, how the ranking of students was generated, and how the grading process occurred. The absence of such information prevented errors from being detected and any reasonable appeals from being raised. Moreover, the practice violated the principle of explainability, which is key in the application of AI.



5. Ethical Issues

5.1 Equality and Distributive Justice:

The functioning of the algorithm only served to further widen socio-economic inequalities, giving more advantage to small class sizes, which were typical of private schools, and disfavoring large class sizes that belonged to public schools. This meant that the students were punished based on their school rather than their knowledge.

5.2 Individual v. Group Judgement:

The primary ethical issue was that students were evaluated based on what other people achieved in the past rather than their own merits. In essence, what mattered in determining the grades was not the student's preparation but rather which individual had attended this particular class five years ago.

5.3 Transparency and Explainability:

The UK government failed to adhere to its own AI ethics principles and overlooked safety and transparency guidelines. The lack of a detailed explanation regarding how the final score was calculated violated the fundamental concept of explainability.

5.4 Lack of “Ethics by Design.”:

When the Ofqual organization rolled out an algorithm across the entire nation without pilot testing, independent audits, or demographic stress tests, it committed an egregious mistake in systems engineering. This should have been identified during the mandatory Data Protection Impact Assessment (DPIA) per GDPR Article 35.

6. Social Issues

6.1 Reproduction of Inequality:

The use of the algorithm further exaggerated the issue of inequality through the limitation of individual scores according to the historical results of a particular educational institution, leaving poor students in state schools subject to punishment through statistics while rich students in private schools were immune.

6.2 Loss of Public Trust in Algorithmic Governance:

Public response to this situation, including demonstrations and numerous protests, demonstrated that decision-making without an explanation and the right to complain significantly decreases trust in government among the population.

6.3 Psychological & Real-world Consequences:

Many students saw their conditional university offers revoked, and legal representation was being offered for those wishing to challenge Ofqual ;this was no mere administrative blunder.

6.4 Distrust in Teachers:

An algorithm was used to replace teachers due to a lack of trust in teachers by those who made policies. “Objective” math was preferable to “subjective” opinions, supposedly to stop grade inflation, yet this revealed an unstated assumption that teachers at state schools cannot be trusted to grade objectively.

7. Legal Analysis

7.1 GDPR Article 22 - The Right against Solely Automated Decision Making:

According to GDPR Article 22, decisions cannot be made about individuals based purely on algorithmic systems because there must be involvement by a real person. In this case, the grades affected university admission procedures; however, no verification was done for individual assessments, and while teachers provided their predictions, the algorithm had full control and could override them.

7.2 Legal Challenges and Judicial Review:

Legal challenges were leveled against the UK government and Ofqual, including discrimination claims from the Equality and Human Rights Commission. These came based on the following provisions

- GDPR Articles 22, 13–15 (automated decision making and right to explain)
- GDPR Article 35 (Data Protection Impact Assessment (DPIA) obligations)
- Equality Act 2010 (indirect discrimination based on race and disability)
- Human Rights Act 1998 / Article 2 Protocol 1 ECHR (right to education)
- Public law principles of procedural fairness (no adequate appeal process)

7.3 Absence of Meaningful Appeal Procedure:

The appeal procedure not being available to individuals violated the principles of data protection and transparency required by law. In accordance with GDPR, individuals have a fundamental right to have someone review and challenge an automated decision; appealing only at school level was not sufficient.

7.4 Non-conducting of the DPIA:

The non-conduct of the DPIA as per Article 35 was a clear infringement of the laws. Since the DPIA needs to be done beforehand for ensuring the fairness of the procedure, the lack of conduct of the same renders the whole installation of the system faulty.

8. Deeper Critical Observations

- The Fallacy of "Objectivity": The application of the algorithm was meant to demonstrate scientific objectivity, but an algorithm cannot be considered objective as everything depends on what data is fed in and how it is optimized.
- Accountability Absence: No one was truly held accountable as Williamson blamed Ofqual, Ofqual blamed the algorithm, and students bore the consequences while decision-makers deflected responsibility.
- Speed Against Safeguards: The algorithm was conceived in June and deployed in August 2020, a two-month window far too short for the audits and fairness testing that a high-stakes national system required.
- Changing Legal Protections: Following Brexit, the UK government considered scrapping GDPR Article 22 to accelerate AI innovation, demonstrating how politically fragile legal safeguards against algorithmic harm can be.

9. Lessons for AI System Engineering

In this case study, several critical engineering ethics guidelines were violated:

- Designing fairness into algorithms: assessing the demographic effects must take place during the design process, not after students have been subjected to the consequences of the algorithm.

- Algorithm explainability: When your algorithm selects where students attend school, people must understand its reasoning about the choice it made.
- Human oversight: Teacher submission of grades is not oversight when the algorithm can override their decision-making process. Oversight means being able to alter the algorithm's decisions.
- Deployment of DPIA: The DPIA must be completed and published before deploying an algorithm.
- Iterated, staged deployment: Do not release an algorithm that is only two months old into a national cohort of students. It should undergo pilot testing first.

10. Analysis of Ethical Issues

The algorithm used for standardization in the 2020 examinations conducted by Ofqual was not a neutral technology but was rather a socially impactful technology that failed the students it had intended to help. There were four key ethical failures in its implementation.

10.1 Biased Historical Data:

Predictions about grades were based almost solely on the performance of the individual schools during previous years. The key issue here is that data gathered from educational establishments does not account for a student's ability independently; instead, it encompasses the effect of funding differences, teacher retention rates, and deprivation indicators. Grades at schools in poorer neighborhoods were traditionally low, and that was a direct reflection of the environment in which children learned, and not the capabilities of the students themselves. As such, the model processed biased historical data, making it the baseline, which then negatively affected the new generation of students. This means that children enrolled in state schools ended up being graded worse while students in private educational facilities suffered no consequences as the historical data about their schools was more robust (Ofqual, 2020). As predicted by O'Neil (2016), this is an example of a feedback loop wherein biased data leads to predictions that seemingly confirm the bias.

10.2 Over-Reliance on Automation:

The teachers have already awarded Centre Assessed Grades based on their continuous observations of their students based on their performance in mock exams, development through coursework, and general behavior during class in two years. The algorithm has overridden about 39% of those assessments (BBC News, 2020). This substitution is based on the belief that a mathematical model trained on data collected from institutions can predict better than the judgment of professionals who know the individual student personally. Not only is this assertion incorrect in the present context but also it shows the technocratic trend of confusing quantity for truth. The use of automation without proper oversight in critical situations will replace human accountability with computational authority and make error detection more difficult.

10.3 Absence of Human Oversight:

It should also be noted that neither students nor teachers had prior knowledge regarding the extent to which the grades assigned by teachers would be overruled. In addition, an available appeals process was not established at the stage of grade releases, and there was no pre-existing documentation or

methodology of the system itself before it became active. Such lack of transparency meant that if there were problems with the system, there was no way to detect and resolve them. It is stated by the EU High-Level Expert Group on AI (European Commission, 2019) that human supervision in automation processes is absolutely necessary for individual rights protection. The Ofqual system completely ignored this requirement and fell apart under its own weight.

10.4 Ignoring Individual Performance:

One of the biggest ethical issues about this case would be the fact that performance of the individual student was of minimal importance in grading him/her. There was no way through which an individual who had outshined their peers in the school could show this performance through the algorithm process. The grading system was more like a prediction about institution performance being used as an individual grade. This is totally unfair because one is supposed to be graded based on his/her performance alone. This algorithm penalised students for attending particular schools which is something that is entirely outside of their control.

11. Responsible AI Practices and Recommendations

The failures above were avoidable and are a result of deliberate decisions that were taken to arrive at the current situation. Recommendations on how to tackle each of the problems mentioned are discussed below.

11.1 Human-In-The-Loop System:

Algorithms need to be employed as supportive tools to make a decision and not as conclusive proof. In cases of such AI systems, teacher evaluations need to be the source of the inputs and the algorithm can only point out abnormalities to the relevant humans without overriding the judgments of the teachers (Shneiderman, 2020).

11.2 Transparency of The Algorithm:

There needs to be an enforceable requirement that requires regulatory authorities to put up algorithmic documentation in the form of model cards or algorithmic impact assessments prior to their deployment. This will entail disclosing the variables, their respective weighting schemes, the limits of the algorithms, and the protective measures available. Such transparency is critical for holding organizations publicly accountable for the performance of such algorithms (Floridi et al., 2018).

11.3 Bias Testing and Equity Audits:

Pre-deployment bias testing needs to assess the performance of the model output across different socioeconomic, geographic, and demographic categories and not only in terms of its effectiveness. The deployment of an equity impact assessment process is needed by law for any application of AI within public services. Post-deployment bias audit is essential to ensure that any new forms of disparities that may arise from the operation of an algorithmic model do not remain unnoticed (Barocas, Hardt and Narayanan, 2023).

11.4 Ethical Governance of AI Systems:

Institutional mechanisms need to put into place the above practices. According to the European AI Act, education falls under the category of high-risk use of artificial intelligence and should have conformity assessments, human intervention measures, and post-market surveillance (European Parliament, 2024). The same should apply to other public organizations around the world.

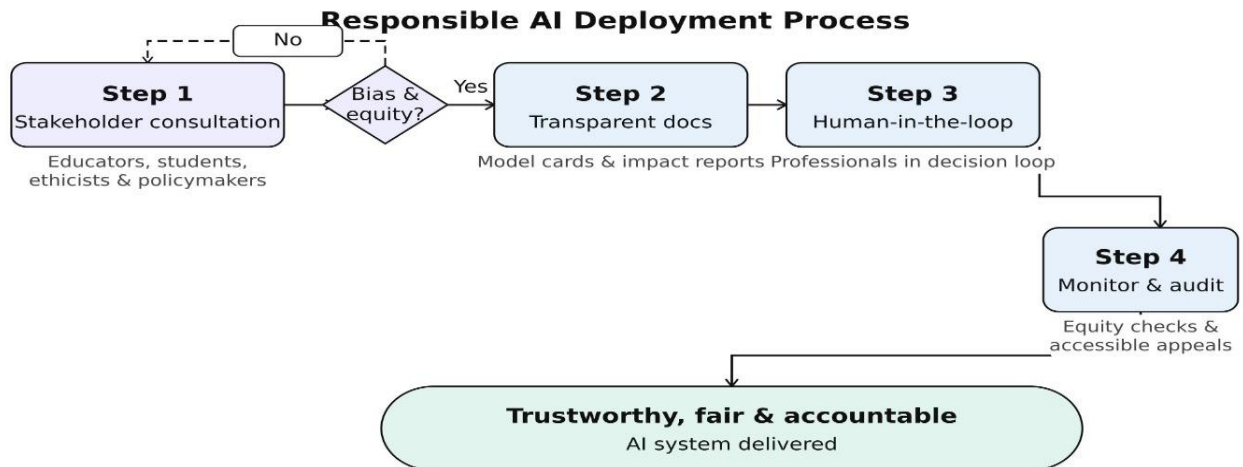


Figure 1: Responsible AI Deployment Process for High-Stakes Public Systems

Conclusion

The Ofqual algorithm scandal of 2020 was not just an error; it was a moral and ethical disaster of the highest order. A system whose goal was to be unbiased ended up institutionalising biases that the algorithms were meant to remove. Students who came from economically disadvantaged families found themselves unfairly disadvantaged not based on what they knew, but rather where they went to school.

From this scenario, it is important to note that automating something does not make it objective and unbiased. The use of an AI system in high-pressure situations with no oversight or transparency will always disadvantage the most vulnerable groups. In this case, it was the decision of the government to reverse its decision that proved once again that the algorithm was flawed.

For engineers of the AI world, the message is clear: fairness must be prioritised right from the beginning.

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